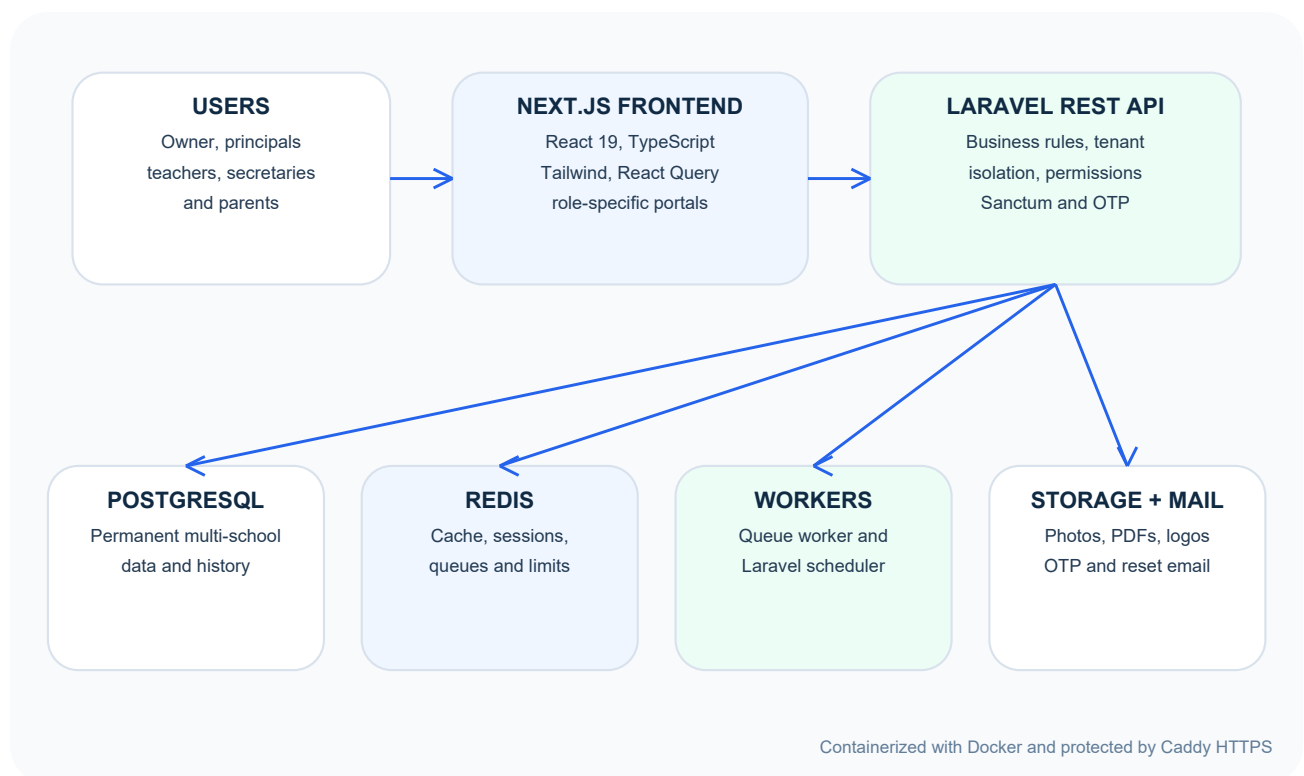


CLYVERO ERP

Architecture, Multi-Service Design and Monthly Deployment Guide

Clyvero ERP is a containerized, multi-service, multi-tenant SaaS platform with a modular-monolith Laravel backend and a three-tier client-server architecture.



Prepared for technical discussions, hosting decisions and future school onboarding.

Reference date: 24 August 2026

1. Executive explanation

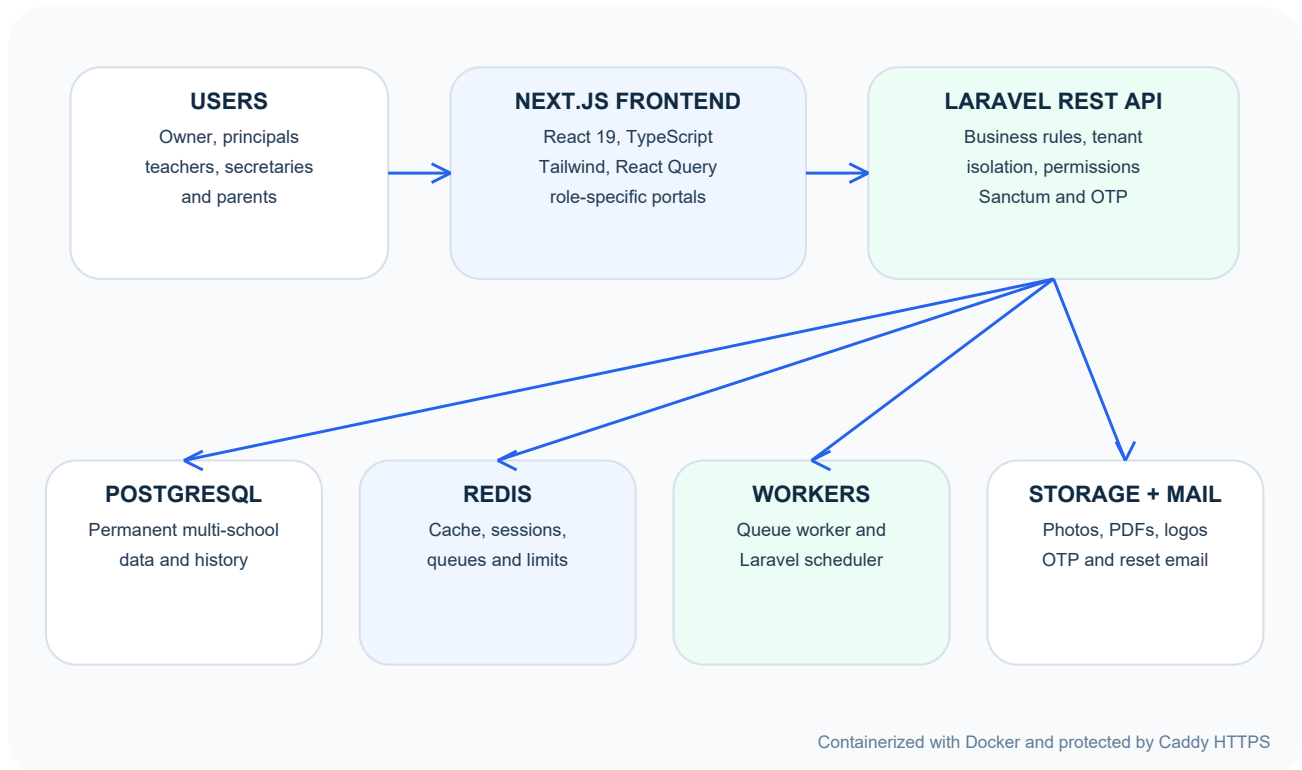
Clyvero ERP is one reusable application that serves multiple schools. Users access a Next.js web interface. The interface sends secure requests to a Laravel REST API. Laravel applies authentication, permissions, school isolation and business rules before reading or changing data in PostgreSQL.

The best short description is: "A containerized, multi-service, multi-tenant SaaS application using a three-tier client-server architecture and a modular-monolith backend."

Why these terms are accurate

Term	Meaning in Clyvero ERP
Multi-service	Several independently running services cooperate: frontend, API, database, Redis, workers, scheduler and proxy.
Not microservices	Student, finance, grades, attendance and academic logic remain modules inside one Laravel application.
Multi-tenant SaaS	One deployed platform serves multiple schools while every school remains logically isolated.
Three-tier	Presentation layer, application/business layer and persistent data layer are separated.
Containerized	Docker provides repeatable environments for development, testing and production.

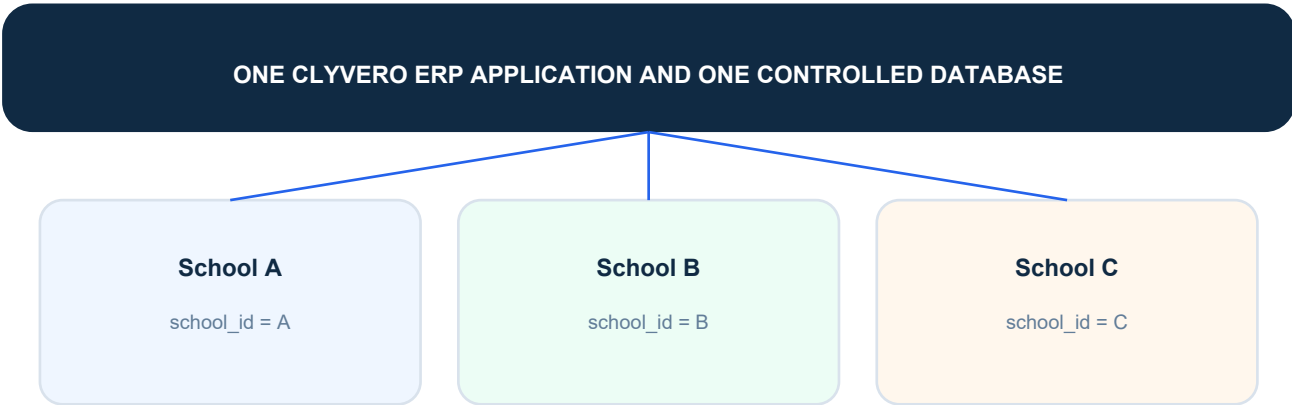
2. Logical application architecture



Layer or service	Technology	Primary responsibility
Web frontend	Next.js 15, React 19, TypeScript, Tailwind	Responsive portals, navigation, forms and API presentation.

Layer or service	Technology	Primary responsibility
Application API	Laravel 12 and PHP 8.4	Validation, workflows, calculations, tenancy, permissions and documents.
Database	PostgreSQL 16	Permanent school, user, academic, finance and historical records.
Cache and queue	Redis 7	Cache, sessions, throttling and background-job transport.
Background execution	Laravel worker and scheduler	Email delivery, queued work and scheduled operations.
Edge and TLS	Caddy	HTTPS, domain routing, compression and security headers.
File layer	Persistent Docker volume	Student photos, school logos, document headers and public files.

3. Multi-tenant data architecture



School-owned records carry a school identifier. Tenant middleware establishes the current school, policies authorize the user, scoped queries restrict access, and PostgreSQL constraints protect relationships. This allows schools to share infrastructure without sharing operational visibility.

Tenant roles

Role	Scope and authority
Platform owner	Global platform role; creates and manages school tenants and principal credentials.
Principal / administrator	Restricted to one school; controls the school's configuration and academic operations.
Secretary	Restricted operational access to permitted student, class, attendance and finance workflows.
Teacher	Sees assigned classes, subjects, timetable, attendance and grade-entry permissions.
Parent	Sees only linked children and the records exposed through the parent portal.

Security model

Laravel Sanctum issues bearer tokens after credential validation. Administrator and platform-owner sign-in can require email OTP. Rate limits, strong password rules, tenant-aware reset links, role policies and tab-local token storage reduce account and cross-role risks. Separate browser tabs can keep different active identities.

4. Owner administration panel

The owner panel is not a second application or database. It is a protected global portal inside the same deployment, available only to the super-administrator role.

Recommended public addresses

Address	Audience
app.clyvero.com	Principals, secretaries, teachers and parents.
owner.clyvero.com	Exclusive platform-owner entry point.
api.clyvero.com	Laravel API used by the web application.

The owner creates a school, supplies its identity and initial principal credentials, and can activate or deactivate the tenant. The principal then manages school-specific fees, subjects, coefficients, academic rules and daily work. Branding changes are stored against the selected school and automatically affect that school's documents only.

Owner account protection

Use a unique password, mandatory email OTP, rate limiting, HTTPS, a private login address, secure recovery procedures and optional Cloudflare access controls. No school administrator can create another platform owner.

5. Initial production deployment

For the first schools, the most economical deployment is one VPS. The services remain logically separate in Docker but share the same machine. This keeps monthly cost and operational complexity low while preserving a migration path.

VPS service	Deployment role
Caddy	Receives HTTPS traffic and routes app, owner and API domains.
Next.js	Serves the public site and every role-specific portal.
Laravel API	Runs business logic and secured REST endpoints.
Queue and scheduler	Runs background and scheduled Laravel tasks.
PostgreSQL	Stores the multi-tenant operational database.
Redis	Supports cache, sessions, rate limiting and queues.
Persistent volume	Survives container recreation and stores uploaded files.

Why the frontend should initially share the VPS

Vercel Hobby is restricted to non-commercial personal use. A sold school platform would require a commercial plan. Hosting Next.js on the same VPS avoids that extra monthly charge. The frontend can still move to another service later.

6. Monthly hosting and scaling strategy

Plan	Resources	Published price*	Use
Hetzner CX23	2 vCPU, 4 GB RAM, 40 GB SSD	EUR 5.49/month	Recommended starting point for the first school.
Hetzner CX33	4 vCPU, 8 GB RAM, 80 GB SSD	EUR 8.49/month	Next step for several schools or sustained load.

*Prices are the official 15 June 2026 prices excluding VAT and Primary IPv4. Provider prices can change. A .com domain remains an annual registrar charge even when server billing is monthly.

Hetzner bills cloud servers hourly with a monthly cap. Starting monthly therefore allows short evaluation periods. Upgrading CPU and RAM is straightforward, but the root disk cannot be reduced. When resizing upward, keep the current disk size when possible so a later downgrade remains available.

Start with 4 GB RAM, not 2 GB. Next.js, Laravel, PostgreSQL, Redis, the worker, scheduler and Caddy all share memory.

7. Capacity and monitoring

Ordinary student, grade, attendance and payment rows are compact. Photos, uploaded documents, generated PDFs, logs and backups will normally consume storage faster than database rows. Capacity decisions must therefore use measurements, not only the student count.

Indicator	Early warning	Action
Disk	70% used	Clean logs/backups, inspect uploads, then add separate storage if required.
Memory	75% sustained or swapping	Inspect containers; upgrade RAM if the demand is genuine.
CPU	70% sustained during normal hours	Profile slow endpoints and queries before adding CPU.
API response	Typical requests consistently above 1 second	Check database indexes, slow queries, cache and network.
Database	Rapid monthly growth	Measure large tables and define retention/archival rules.
Queue	Jobs remain pending	Inspect failed jobs and add worker capacity.

Backup policy

Create nightly PostgreSQL and upload-volume backups, verify checksums, copy them to encrypted off-site storage and perform a restoration rehearsal. A backup that has never been restored is not yet proven.

8. Production launch sequence

Step	Required outcome
1	Purchase the domain and create a private source repository.
2	Create the Hetzner VPS, SSH key and firewall rules.
3	Install Docker and deploy the tagged Clyvero release.
4	Configure production secrets, SMTP and app/owner/API DNS records.
5	Run the production validator, Docker builds, migrations and optimization.
6	Create or rotate the platform-owner account and verify OTP.
7	Test every role and critical school workflow on a staging tenant.
8	Create and restore a full backup before admitting real school data.
9	Monitor the first month and scale only from measured evidence.

9. Answers for technical discussions

Is the application multi-service?

Yes. Next.js, Laravel, PostgreSQL, Redis, queue workers, the scheduler and Caddy run as separate services. The business backend is still a modular monolith, not a collection of independent microservices.

Can one deployment serve several schools?

Yes. Clyvero uses a multi-tenant model. Each school receives an isolated tenant inside the shared platform, and its users, branding and operational records remain scoped to that school.

Can the system scale later?

Yes. The first deployment keeps services on one VPS. Later, the database, uploads, workers or frontend can move to separate infrastructure without replacing the school-management domain model.

Recommended elevator pitch

"Clyvero ERP is a secure, bilingual, multi-tenant school-management SaaS platform. It uses a Next.js frontend, a Laravel REST API, PostgreSQL and Redis, coordinated as containerized services. One platform serves multiple schools while tenant controls isolate every school's users, branding and data."

10. Sources

Hetzner Cloud billing FAQ: <https://docs.hetzner.com/cloud/billing/faq/>

Hetzner resizing and disk downgrade FAQ: <https://docs.hetzner.com/cloud/servers/faq/>

Hetzner CX resource specifications: <https://www.hetzner.com/cloud/cost-optimized>

Hetzner price adjustment effective 15 June 2026:

<https://docs.hetzner.com/general/infrastructure-and-availability/price-adjustment/>

Vercel Hobby plan and commercial-use restriction: <https://vercel.com/docs/plans/hobby>

Vercel pricing: <https://vercel.com/pricing>

Contabo plan changes and downgrade procedure: [Official Contabo support article](#)

Pricing and provider rules are time-sensitive. Reconfirm them immediately before purchase. This document describes the recommended technical direction and does not replace a provider's final quotation or terms.